Kadi-APY Workshop

Automating everyday data management tasks in Kadi4Mat with Python and the command line

DATE AND TIME
Friday, 13 November 2026
14:00-16:00 (CET)

FORMAT
Online, hands-on
in VS Code (GitHub Codespaces)

About the workshop

Kadi-APY is the Python library and command line tool of Kadi4Mat. It allows routine steps of the web interface, such as creating records, uploading files, adding metadata and searching, to be scripted. This is useful whenever the same steps are repeated for many files, for example after a measurement series.

We apply it to tensile tests of three steel specimens (synthetic data): the result files are uploaded, calculated values such as the yield strength are stored as metadata, and the data is found and plotted again. The same steps apply to other instruments and file types. No prior experience with Kadi-APY or Python is required.

Schedule

Each part consists of a complete example, a reference of the functions used with links to the Kadi-APY documentation, and an exercise.

TIME	BLOCK	CONTENT
14:00-14:10	**Welcome and demonstration**	Kadi4Mat terms and a live demo: three specimens uploaded with one script.
14:10-14:20	**Part 1: Kadi setup check, Kadi-APY documentation**	Kadi4Mat access; finding and reading functions in the Kadi-APY documentation.
14:20-14:35	**Part 2: Command line**	Record, file, metadata and tags with single commands; for quick one-off steps.
14:35-14:55	**Part 3: Python**	The same steps as a script; the basis for processing many files.
14:55-15:00	Break	
15:00-15:20	**Part 4: Automation**	One script for a folder of result files: records with calculated values as metadata, files, grouped in a collection, and links to related records.
15:20-15:30	**Part 5: Search, download, plot**	Find records by tag, plot their metadata, download the result files and plot the measured data.
15:30-16:00	**Discussion**	Your own use cases, which functions would automate them, open questions.

Preparation and material

Before the workshop (about 20 minutes): create a personal access token in Kadi4Mat (kadi4mat.iam.kit.edu, Settings > Access tokens), open the repository in GitHub Codespaces, install kadi-apy and set host and token with `kadi-apy config`, as on your own computer. A step-by-step checklist is in the participant guide. Google Colab is available as a backup.

Open in Codespaces | github.com/msishuvo/thematic-workshop-Kadi-APY | kadi-apy.readthedocs.io